Water Quality Prediction and Pattern Recognition by Using Machine Learning and Federated Learning

by

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Abstract

Water is one of the most precious resources in the world, and monitoring its quality is fundamental to human health and sustainable development. Modern machine learning algorithms in both centralized and decentralized settings have made these predictions more accurate and efficient. Using federated learning over centralized models can achieve comparable forecasts with fewer resources, less communication overhead and lighter models. The forecasts these models give may vary depending on feature correlations, target variables, dataset composition, and sample size . This study compares , analyzes and forecasts the various trends and patterns of a global dataset spanning over 2.8 million water quality records (1940–2023) in both central and decentralized model environments using XGBoost, Random Forest Regression, SVR and for FL uses FedAvg , FedOpt, FedProx models. The results establish that non-linear models significantly outperform linear models like SVR due to nonlinearity of the dataset. The FL models also can perform quite similarly to non-linear models in predictive accuracy and convergence behavior. Federated Learning can also reduce cost on resources and on communication overhead, training time, storage cost . These findings show that contrary to traditional assumptions, FL models can get comparable results like centralized ML models and non-linear machine learning models can predict real world water quality data quite accurately. The study demonstrates the feasibility of scalable, reliable and cost-effective solutions to global water resource management using federated and centralized learning approaches..

Keywords: Federated Learning, Machine Learning, XGBoost, Random Forest Regression, SVR, FedAvg, FedOpt, FedProx

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Chapter 1

Introduction

1.1 Background

Detecting water quality is essential for agriculture and human consumption yet heavy contamination from garbage disposal, industrial waste and unsafe sewage treatment has left much of the drinkable water sources unsafe. Traditional methods to observe such water are costly, labor-intensive, limited in scalability , time-consuming and resource-intensive to predict the water quality. Using machine learning models in such cases can give more accurate results and prediction while also finding out any underlying patterns. And federated learning can help in these models with a decentralized approach that reduces both communication overhead and required computation power.

1.2 Rational of the Study or Motivation

Water quality directly affects the health of a populace. In many areas of the world people still lack access to safe drinkable water sources. Providing information to these affected people so that these population centres can easily access clean drinking water motivates this study. With openly available large datasets, machine learning provides the unique opportunity to develop predictive models that can easily assess the quality and the cleanliness of any water source. For the real world use in these resource-constrained regions federated learning was used as it can provide cost-effective, efficient ,reliable solutions to the challenges of water monitoring in resource-constrained regions with low storage cost and communication overhead.

1.3 Problem Statement

Access to clean water still remains a challenge to many parts of the world. Machine learning offers the opportunity to accurately and efficiently monitor and predict water quality using real world large datasets. ML can identify many complex underlying patterns and anomalies [13]. But centralized learning also has its challenges to overcome. Centralized approaches require high computational resources for these models with scalability issues. Federated learning provides a solution to these problems with decentralized models. The effectiveness of these decentralized models against computationally intensive centralized models remains underexplored. The lack of systematic evaluation and the accuracy of federated models and forecasts that FL delivers is the critical gap that this research targets to address.

1.4 Objective

The primary objectives of this study are given below:

1. **Accurately predicting water quality :**
   - To preprocess large-scale real world datasets.
   - To predict water quality based on real world data.
2. **Comparing different FL and ML models:**
   - To implement and evaluate between different ML(XGBoost, SVR,Random Forest Regression) and FL(FedAvg, FedProx, FedOpt) models .
   - To indentify the best performing model for real world datasets.
3. **Future Water quality forecast:**
   - To predict the future water quality of water sources at current rate of pollutions.
   - To analyze global water quality trends for long term clean sustainable goals.

These objectives collectively aim to create a robust, efficient, and cost-effective system capable of handling high data volumes without compromising predictive performance of any models.

1.5 Methodology in Brief

The study follows a systemic approach to evaluate the performances of different sorts of centralized and decentralized models.
The source dataset is initially cleaned and preprocessed for the models and checked for any sort of null, wrong and corrupted data. This data then goes through feature scaling and normalization for the models.
The cleaned dataset is used by different ML(XGBoost, SVR,RandomForest Regression) and FL(FedAvg, FedProx, FedOpt) models. All models were trained after identical situations in 2 devices separately to ensure reliable and robust results. The centralized models were run on a single server whereas the decentralized models were separated in different clients according to their country.
Prediction accuracy was based on R^2 score, Mean Absolute Error (MAE), and Root Mean Square Error (RMSE) with the system evaluation being assessed using training time , logloss while forecasting accuracy was analyzed using time series and error rates.

The founding idea of this research was guided by the paper [13]

1.6 Scopes and Challenges

Scopes:

This study focuses on the application of centralized and decentralized models on real world water quality data. Here the centralized models used are XG-Boost, Random Forest Regression for non-linear learning and SVR for linear learning. For federated learning we use FedAvg, Fedopt, FedProx. The research evaluates the models with predictive accuracy, R2 score, RMSE, MAE, training time , logloss and storage cost. The purpose of this study was to develop a stable water predicting model using FL for resource constrained areas because centralized learning would be inapplicable here.

Challenges:

Despite the broad scope, the study faced quite a lot of hurdles. For starters the global dataset was large in scale with inconsistencies and varying measuring standards which required a lot of time and computational power to preprocess. The next step, forecasting is inherently uncertain due to the presence of many contributing factors such as sudden environmental tragedies, unknown pollutants. Centralized learning does provide a stable accuracy with the caveat of being resource intensive whereas the decentralized models do show potential in this regard but with a slower convergence rate. The key takeaway here was to balance accuracy with efficiency for real world application. .

Chapter 2

Literature Review

2.1 Preliminaries

A major limitation in this domain has been the lack of large-scale, high-quality datasets. This challenge was addressed by [13], who published a comprehensive dataset of surface water quality measurements spanning 1940–2023. This resource enables both empirical research and the application of advanced ML models, supporting trend analysis and cross-regional comparisons over a long temporal scale.

Now, on the data side [13] shows the unique foundation that ML lays when doing research on large dataset covering water quality from 1940 to 2023. Previous work in the domain from [11], [9], [1] demonstrated how supervised machine learning ML uses the hyperparameters and hyperparameter optimization to improve prediction accuracy but it's often used on smaller or regional databases. These previous studies show the need for scalable approaches when handling enormous global water quality data.

From computation resource point of view , federated and edge computing studies from[10] , [10] , [7] ,[8], [4] , [5] and [2] gives proper guidance for secure and efficient strategy that are also prone to disability from latency related issues. Combined they show the advancement in FL methods using all the way from FedOpt optimization to client selection and privacy-preserving frameworks which can be applied in these water quality index prediction scenarios. Finally [3] shows how 1D CNN architecture can easily be used on time series signals that can process water quality measurements.

2.2 Summary of Key Findings

The highlights of these literatures provide three interconnected insights. First, data availability and quality are of utmost importance : while comprehensive datasets [13] open new possibilities of consistency and integration across multiple decades remain a challenge. Then scalability and cost-efficiency reappeared on multiple papers. 2 industry reports from [12] and academic survey from [6] emphasize the need for careful calculation on cloud costs. Lastly, though most FL research target much more advanced domains like vehicular networks or smart grids the core principle of resource optimization [8] and secure collaboration [5] are directly correlated to environmental data monitoring like water quality.

Overall, the literature suggests that prediction of effective larger dataset water quality prediction requires a convergence of **efficient ML models with optimized federated deployment on cloud infrastructures**. The weaknesses of such builds are limited generalizability of models, outdated usage of cloud statistics, and domain-specific FL studies which exposes a gap this research aims to fill by building scalable, cost-effective , and accurate water quality prediction pipelines with forecasts through both advanced machine and federated learning.

Chapter 3

Dataset

3.1 Dataset overview

For this research two data sets were used:

The first dataset was named [combined data] 2.0 . Which served as the primary dataset for our research. All the running models and algorithms were run and tested on this dataset. This dataset had the needed scale and diversity for all the debugging , benchmarking and fine tuning for the project.

The second dataset is named [China water pollution data] . This data was intentionally kept out of the training round to ensure a fair, unbiased evaluation of all the models.

By using this 2 dataset model this study can validate and refine its models and its results better. Thus, leading to a more ronust model and relaiblilty for the result.

3.2 Dataset Preprocessing

This study uses 2 datasets. The first dataset was extensively used and tested for benchmarking models. The second model was only used for testing as in stimulating real world data.

Preprocessing were done on these datasets using OpenRefine to ensure reliability , efficiency and reusability. Text fields such as (Country, Water body, Area) were all trimmed and standardized. Then a unique key was generated to look for duplicates and was dropped afterwards and columns were renamed for clarity (pH (ph units) $\rightarrow$ pH; Temperature (cel) $\rightarrow$ Temperature (°C)). Numerical attributes such as pH, BOD, DO,Temperature, Ammonia, Orthophosphate, Nitrogen, and Nitrate were all converted to numerical formats. The dates were in dd-mm-yyyy format but they were converted to the same format again just to get rid of any anomaly. Impossible values were also removed such as pH outside of (0-14) and also rare or analmalous values were also removed such as temperatures above 100.

After the preprocessing the dataset was standardized, normalized and had validated numeric records. Then the data was divided for ML and then by country again for FL to simulate real-world decentralized datasets

❖ Total rows: 2827977

❖ Unique values per column:

	Unique Count
Country	5
Area	61689
Waterbody Type	12
Date	23876
Ammonia (mg/l)	13987
Biochemical Oxygen Demand (mg/l)	4717
Dissolved Oxygen (mg/l)	16253
Orthophosphate (mg/l)	10596
pH (ph units)	4726
Temperature (cel)	11536
Nitrogen (mg/l)	16232
Nitrate (mg/l)	10823
CCME_Values	220872
CCME_WQI	5

Figure 3.1: Cleaned Dataset info

Chapter 4

Proposed Methodology

4.1 Centralized models

In the centralized model all the data used the dataset were first cleaned, preprocessed and stored on a single server without any partitioning on a single server. The data was stored, normalized and temporarily sorted to preserve the chronological consistency. The feature set included were Ammonia, Biochemical Oxygen Demand (BOD), Dissolved Oxygen (DO), Orthophosphate, pH, Temperature, Nitrogen, and Nitrate, with CCME Values used as the regression target and CCME WQI as the categorical target for classification. For a robust evaluation the first dataset were divided into training ('64%'), validation (16%), and test (20%).

In this setting 3 models were used:

SVR:
SVR was used to find the simple linear pattern and dependencies between water quality features and CCME indicators. Here LinearSVC was used for classification with balanced class weighting to counter class imbalance. This baseline model gave the primary insight into the limitation of linear models for complex , non-linear real world data.

XGBoost:
A slow gradient boosting with GPU acceleration was to handle the large dataset. It was used for both regression and classification. This model employed early stopping,subsampling, and regularization to balance between overfitting and accuracy. The model were run on multiple subset of features to create a robust and reliable result.

Random forest regression:
Another non linear model to find the patterns with the help of ensemble decision trees. Cross validation was applied in this model to evaluate consistency and to create a stable with robust accuracy.

Evaluation metrics for the centralized models included R^2, MAE, RMSE (regression) and Accuracy, F1, Balanced Accuracy, Log Loss (classification), alongside training time. For further analysis and also for prediction Prophet was used for 5 year prediction with the help of the ML model.

This centralized model compared linear(SVR) and Non-linear(XGBoost, Random Forest) Models where SVR performed to quite poorly against other non-linear models.

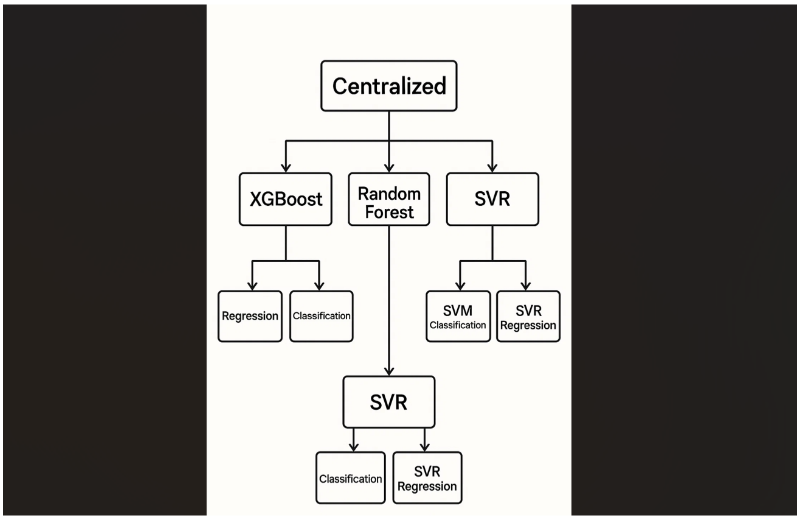

Figure 4.1: Centralized Models Diagram

4.2 Decentralized models

To mimic the real world decentralized dataset, federated learning models were chosen. Here each client was partitioned according to their country for an unique geographical distribution of water quality records. Local training was done on each client node while only the update was shared with the global model with pytorch and flower. The features used were Ammonia, BOD, DO, Orthophosphate, pH, Temperature, Nitrogen, and Nitrate, with CCME Values (regression) and CCME$_WQI(classification)astargets.$

The following federated models were tested:

FedAvg:
The baseline algorithm where we use local cnn to train the client data and then each client sends an update to the global model. The server which hosts the global model then averages the parameters received for the global model. This method proved stable but struggles with heterogenous client distribution.
FedProx:
This algorithm is a mutation of FedAvg where the algorithm employs a proximal term in each and every local training to penalize large deviations from the global weights following the global model. Thus ensuring robustness even in highly imbalanced local data.
FedOpt:
In this method an advanced optimized framework is used where the global server

uses adaptive optimizers (FedAdam) instead of regular averaging. As the backbone of this model a residual MLP with dropout regularization is used. This model severely improves the convergence and yields a higher F1 score.
Every client together created the optimized regression (CCME$_{V}alues)andclassification(CCME_{W}QI$
$levelmetricssuchasconvergencespeedandcommunicationoverhead.$

Forecasting used prophet through time series for future 5 year water quality prediction just as centralized models for predictive modeling with long-term water quality management.

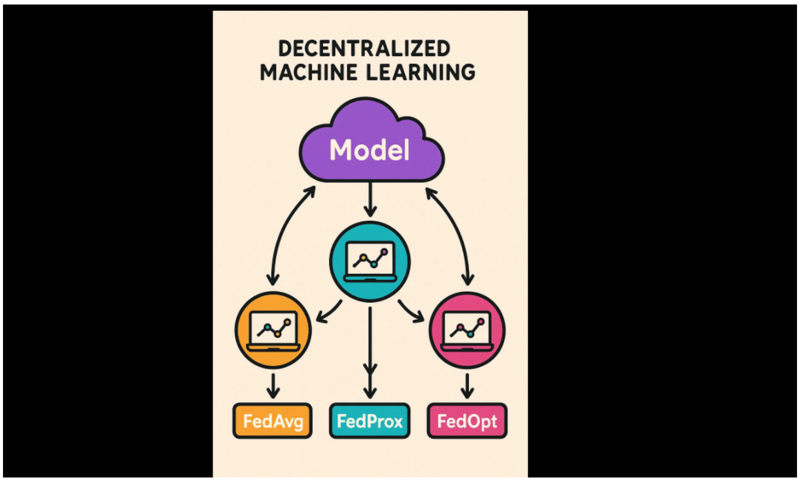

Figure 4.2: Federated Learning Diagram

Chapter 5

Result Analysis

5.1 Performance Evaluation

This section evaluates both the regression and the classification metrics of all the **centralized** and **decentralized models**. Figures are given sequentially for each model, followed by comparative analyses.
Feature importance analysis showed that Ammonia, Orthophosphate, and Biochemical Oxygen Demand (BOD) were the most influential factors when predicting the water quality . Temperature, nitrogen, and pH followed then in importance but also contributed significantly.

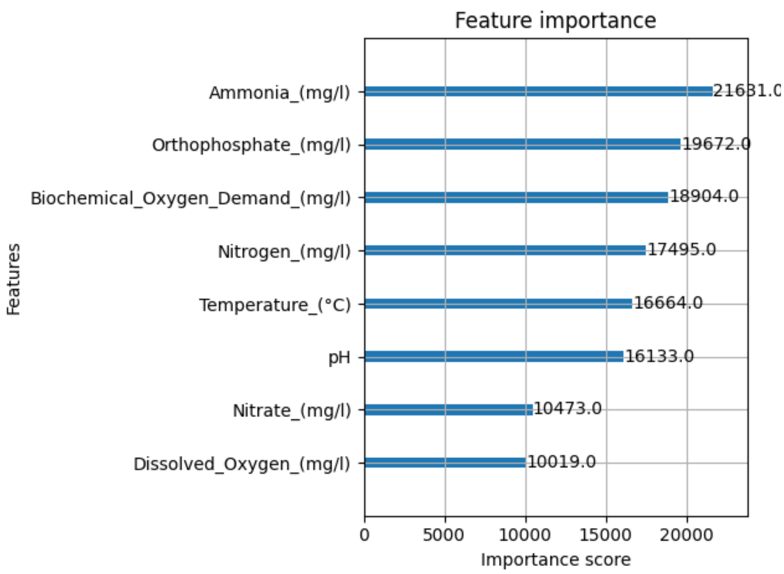

Figure 5.1: Classification Feature Importance

XGBoost (Centralized):
XgBoost Performs the best out of all the models in this paper. Regression was working well with R^2 close to 0.99 and low MAE/RMSE. Classification performance was also excellent, with accuracy above 99% which translates to also good F1 score with low logloss. To make the model more robust then it was tried with less correlating

features which gave a slightly lower result but still could work with the reduced features.

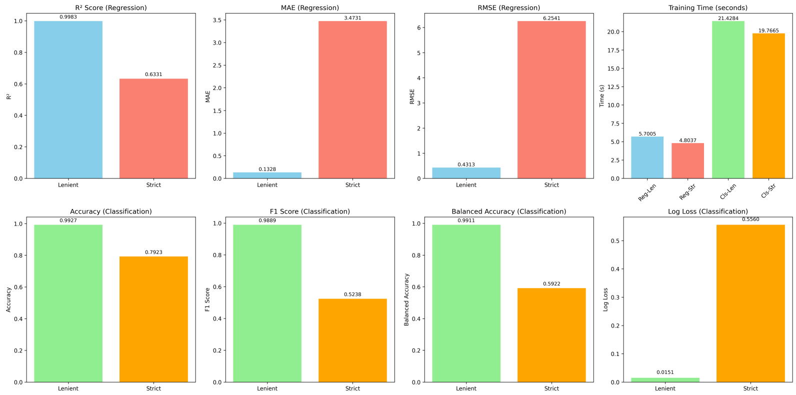

Figure 5.2: XGBoost

Linear SVR (Centralized):
LinearSVR was comparatively weaker compared to the other non-linear models. Regression R^2 was unstable, and error values (MAE and RMSE) were high. Accuracy in classification and balanced classification both were modest, which means the model was struggling when capturing non-linear patterns. But it did train faster, making it more suitable for low resource situations.

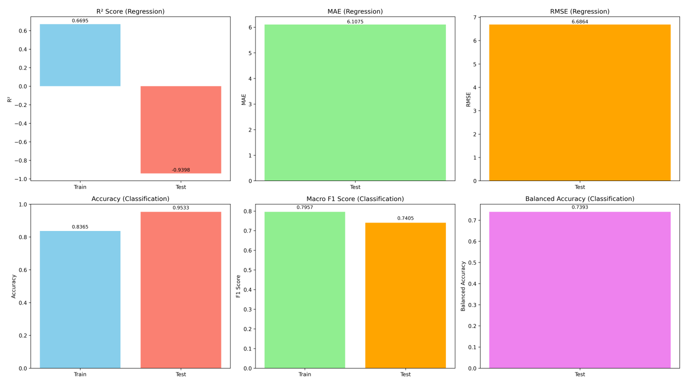

Figure 5.3: Linear SVR

Random Forest (Centralized):
Random Forest performed excellently a near perfect result in lenient situation with all the features. To test this perfect model further top 3 correlating features were taken away which reduced the regression R^2 while also downing the classification metrics. To stress this model even further in stricter run the next 3 correlating features were also taken away while evaluating thus truly testing the model in hard situations. But Random Forest still gave robust and reliable results which were also interpretable.

FedAvg (CNN):

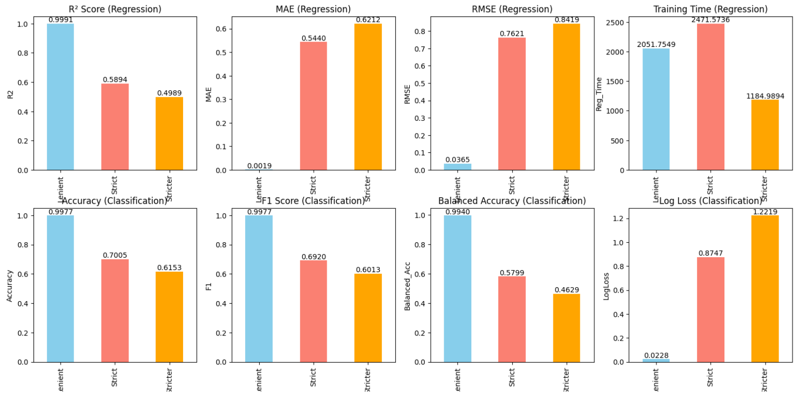

Figure 5.4: Random Forest

The FedAvg model achieved a final classification accuracy of 80% then the regression mrics (MAE 98.5, MSE 9727.6) followed by similar scales as other federated approaches, but they did not improve prediction. This result indicates that FedAvg model struggled in federated CNN training for water quality prediction. The client-level analysis showed variability across different client participants, but the aggregation still provided stability in the global model.

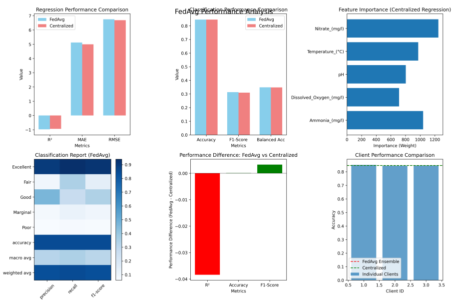

Figure 5.5: FedAvg

FedProx (CNN):

FedProx model achieved about 80% in classification accuracy. Then regression R^2 remained unstable(went negative), convergence plot revealed that while classification accuracy stayed somewhat stable across rounds, regression R^2 failed to improve, highlighting a limitation in FedProx when handling the regression tasks. Data parameter sensitivity analysis suggested that when tuning the proximal parameter μ could very slightly improve the performance of the model.
Federated Methods: Final Accuracy

The model comparison across all the federated strategies shows a clear performance hierarchy, though some are still below the centralized training baselines:

FedAvg (CNN) achieved the best classification accuracy at around 80–84%, which is closely following the centralized model.

FedProx (CNN) reached around 70–78% accuracy, thus showing a moderate stabil-

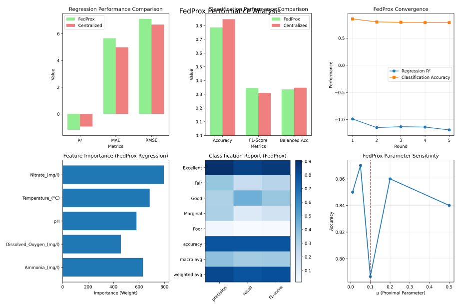

Figure 5.6: FedProx

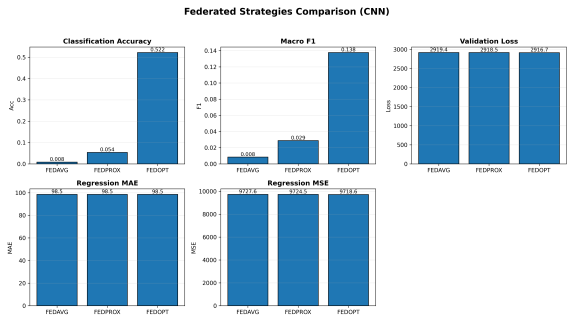

Figure 5.7: Combined FL models

ity but slightly weaker convergence when compared to FedAvg.

FedOpt/FedAdam achieved around 50–52% accuracy, performing significantly better than the very low values that was seen in earlier trials but still trailing a lot behind FedAvg and FedProx.

On the regression side of models , none of the methods achieved strong results. All three reported negative or unstable R^2 scores and consistently high error values (MAE 98.5, MSE 9718–9728). The FedProx and the FedOpt showed slightly lower MSE than the FedAvg, but the improvement was negligible when compared . Overall, these findings highlight that while FedAvg model remains the most reliable approach for classification accuracy (80%), FedProx model provides acceptable but weaker stability (70%), and the FedOpt model trails behind at 50%. In regression tasks, however, all federated methods really struggled to generalize, thus suggesting further optimization and algorithmic adjustments are required for practical deployment.

5.2 Analysis of Design Solutions

The design solutions explored in this research consisted of different centralized machine learning models (Example:XGBoost, Random Forest, Linear SVR) and many federated learning algorithms (FedAvg, FedProx, FedOpt). Centralized solutions demonstrated very high accuracy , reliability and robustness.

XGBoost achieved the strongest performance among all , with regression R^2 close to 0.99, low MAE/RMSE, and classification accuracy above 99% (Figure 5.2). All these came down a bit when features were reduced. Performed good on Non-linear patterns

Random Forest was very effective in lenient settings but performance degraded under stricter evaluation (Figure 5.4). It could also find good non-linear patterns.

Linear SVR trained faster but suffered from low regression accuracy and modest classification results (Figure 5.3). It was good on linear patterns but performed worse on non-linear conditions.

Federated solutions offered privacy-preserving computation but at the cost of reduced accuracy.

FedAvg achieved very good 80% accuracy and displayed excellent convergence trends (Figure 5.5) among all the FL models.

FedProx struggled, with lower 70-78% accuracy and weaker regression performance (Figure 5.6) when compared to FedAvg.

FedOpt underperformed compared to others with 50% accuracy, but it can be blamed on the pipeline as it was run on a script on the last where other drew the resources most. (Figure 5.7).

Overall, all the evaluation criteria—accuracy, efficiency, feasibility, and reliability—showed that centralized models are slightly more stronger for prediction, while federated learning introduces scalability and privacy trade-offs but they can be cheaper to implement on the real world.

5.3 Final Design Adjustments

Based on all the evaluation results:
Centralized adjustments:

XGBoost was selected as the primary model among all the centralized models due to its balance of predictive power and efficiency while also retraining efficiency.

Feature reduction experiments showed and confirmed robustness, as the model retained performance even with fewer correlated variables (Figure 5.2).

Federated adjustments:

FedAvg was chosen as the most viable and cheaper model for federated method, since it consistently outperformed FedProx and FedOpt (Figure 5.5).

Adjustments for FL models included exploring communication-efficient aggregation and parameter tuning to improve convergence for higher accuracy.

FedProx required tuning of the proximal parameter μ (Figure 5.6), while FedOpt was identified as promising for tabular CNN extensions that worked great in pipelines.

5.4 Statistical Analysis

The statistical results of all the models used classification and regression to assess the performance of all the models employed.
Regression metrics: R^2, MAE, RMSE were used to provide insights into the predictive accuracy and error distribution of all the models.

Classification metrics: Accuracy, F1 score, Balanced Accuracy, and Log Loss were used to assess the quality of categorical predictions for each model.

Figures 5.2, 5.3, and 5.4 highlight the various centralized performance across these metrics. Federated results recieved from (Figures 5.5, 5.6, 5.7) emphasize on the reduced accuracy but confirm the validity of these distributed training setups for water quality prediction.
Feature importance analysis (Figure 5.1) confirmed that Ammonia, Orthophosphate, and BOD are statistically the most relevant predictors of water quality that is also the most correlated to the water quality. Removing them have an adverse effect on the whole model.

5.5 Comparisons and Relationships

A comparison across all models in the study shows: Centralized superiority:

XGBoost outperformed all models, followed by Random Forest, with Linear SVR as the weakest without the federated learning models.

Centralized approaches consistently and efficiently produce higher accuracy, lower errors, and greater stability but require higher upfront capital to employ in the real world.

Federated learning trade-offs:

FedAvg > FedOpt > FedProx in terms of accuracy for real world data (Figure 5.2,5.3,5.4).

Despite slightly lower performance, FL methods demonstrated practical value for low cost computations in distributed datasets without the need for resource intensive computation systems..

Relationship with existing works:

The strong centralized performance of XGBoost model aligns with the state-of-the-art literature on tabular and structured datasets.

The reduced performance of FL models reflects the current research challenges in federated learning—particularly non-i.i.d. client data and limited communication efficiency. But they become competitive when considering their low cost approach.

5.6 Discussions

The results show that the usage of centralized models, particularly **XGBoost**, delivers superior accuracy and stability. XGBoost still remains practical even when trained on reduced features, meaning it could support **low-cost water monitoring stations** without a significant loss on predictive power.

On the other hand, **federated learning (FL)** is more difficult to implement on code properly due to a variety of issues such as unstable convergence, client heterogeneity, and parameter sensitivity. While accuracy was slightly lower across FedAvg, FedProx, and FedOpt, FL offers a **cheaper and more privacy-friendly alternative** in very resource-constrained or money-strained settings, since it avoids the need for large expensive, capital intensive centralized infrastructure.

In practice, centralized models show that they are better for accuracy and reliability even with reduced features, while FL provides a cost-efficient option when budgets are limited, privacy is a concern and data decentralization is required.

Chapter 6

Conclusion

6.1 Summary of Findings

This study demonstrates that the usage of centralized machine learning models, particularly XGBoost, provides the highest predictive accuracy and stability for water quality forecasting and prediction . XGBoost remains effective even with reduced feature sets, making it a practical choice for deployment in low-cost monitoring stations. Random Forest and Linear SVR also show really competitive performance, but with limitations in either scalability or interpretaibility compared to XGBoost.

Now ,in federated learning models such as FedAvg, FedOpt, and FedProx achieve and get slightly lower accuracy. But FL exhibit various unstable convergence due to client heterogeneity and extreme parameter sensitivity. Nonetheless, FL offers some unique advantages in cost-efficiency, privacy preservation, and decentralized data handling, making it a valuable and practical alternative for resource-constrained or data-sensitive environments where centralized infrastructure is not feasible.

Overall, the findings suggest that using centralized models are best suited for high-performance and in situations where high reliability in water quality prediction matters the most , while federated learning represents a promising direction for future research in enabling affordable, privacy-aware solutions in financially strained settings for millions who can't access clean drinking water.

6.2 Recommendations for Future Work

While the current study demonstrates the relative strengths of centralized and federated approaches for water quality prediction, there remains several opportunities for improvement and extension:

- **Enhanced Data Coverage:** Using various more complex datasets and models for further usage such meteorological data, satellite dataset and ground water datasets to make the models more accurate and applicable to the real world.

- **Feature Optimization:** Optimizing computational and monitoring costs and further stabilize the feature selection method to make the models cheaper and easier to deploy in the real world.

- **Federated Learning Stability:** Further research should focus on stable and more efficient implementations of various FL models to improve convergence and reliability
- **Cost-Aware Deployments:** Developing lightweight model architectures best suited for low-power edge IoT devices and rural water stations can for real-world deployment that can make impact in practical resource-constrained settings.

These directions highlight future possibilities for combining high-accuracy centralized models with cost-effective and privacy-preserving methods in federated learning, ultimately enabling more inclusive and resilient water quality monitoring systems that can help millions more people for a much healthier life.

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